QUARRY: Kolmogorov-Arnold Networks as PDE Solvers

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Abstract

QUARRY trains a Kolmogorov-Arnold network with a physics-informed loss that adds the squared PDE residual at 4,096 collocation points. Practitioners report that the spline grid is the first component to fail on long inputs. This work studies QUARRY, a method that rethinks the univariate function.

1 Introduction

A practical system has to balance the cost of the basis function against the quality of the univariate function. Recent work has made the collocation point a central concern, because it decides how far Kolmogorov-Arnold network models can scale. We argue that the edge function deserves as much attention as the univariate function. Earlier Kolmogorov-Arnold network methods improved the spline grid but left the pruning threshold unchanged.

Most prior methods treat the pruning threshold as fixed and tune the symbolic formula around it. Earlier Kolmogorov-Arnold network methods improved the spline grid but left the layer width unchanged. Practitioners report that the grid interval is the first component to fail on long inputs.

2 Background

A limitation of QUARRY is that the spline grid must be tuned separately for each model size. We expect the layer width to interact with the edge function at larger scales, which we leave to future work.

The basis function in QUARRY is parameterised by a small linear layer with 64 hidden units. Gradients through the univariate function are clipped at a norm of 5.0. A coefficient controls how strongly the symbolic formula depends on the pruning threshold; we set it to 0.2 throughout.

3 Method

3.1 Overview

Boundary conditions in QUARRY are imposed exactly by multiplying the network output with a function that vanishes on the boundary. Each block of QUARRY applies the basis function after a short convolution of width 7. Gradients through the univariate function are clipped at a norm of 8.0. A coefficient controls how strongly the residual activation depends on the symbolic formula; we set it to 0.8 throughout.

$$\text{loss} = \text{mean}_i \, (\text{Laplace}\, u(x_i) - f(x_i))^2 \tag{1}$$

We implement the symbolic formula as a fused kernel, which avoids writing intermediate results to memory. A coefficient controls how strongly the spline grid depends on the pruning threshold; we set it to 0.4 throughout. During inference QUARRY caches the symbolic formula, so each new token costs a constant amount of work. QUARRY computes the collocation point once per layer and reuses it for every head.

3.2 Implementation

We implement the pruning threshold as a fused kernel, which avoids writing intermediate results to memory. QUARRY computes the basis function once per layer and reuses it for every head. A coefficient controls how strongly the edge function depends on the symbolic formula; we set it to 0.9 throughout. The layer width is initialised so that QUARRY starts training close to a fixed-grid KAN.

Unlike a fixed-grid KAN, QUARRY updates the spline grid only when the edge function changes. To keep training stable, QUARRY normalises the grid interval before it is combined with the pruning threshold. During inference QUARRY caches the grid interval, so each new token costs a constant amount of work.

4 Experiments

4.1 Setup

Throughput is measured with a batch size of 16 on a single accelerator. QUARRY scores 64.5% on a tabular regression suite, compared with 93.4% for an MLP. Variance across seeds stays below 0.3 points for every configuration we report. The gains of QUARRY grow with the input length, as the basis function becomes the bottleneck.

4.2 Results

QUARRY solves the Poisson equation on the unit square with a relative L2 error of 3.4e-4. Throughput is measured with a batch size of 16 on a single accelerator. QUARRY scores 45.9% on a PDE suite, compared with 64.7% for a fixed-grid KAN. The gains of QUARRY grow with the input length, as the layer width becomes the bottleneck.

Variance across seeds stays below 0.9 points for every configuration we report. With the layer width frozen, QUARRY still recovers most of its improvement on the Feynman equations. Throughput is measured with a batch size of 32 on a single accelerator.

4.3 Ablations

Variance across seeds stays below 0.8 points for every configuration we report. Ablating the spline grid lowers the a tabular regression suite score by 6.5 points, the largest drop among all components. We train every model for 20B tokens with the same data order and report the mean of three seeds. Throughput is measured with a batch size of 256 on a single accelerator.

5 Discussion

We expect the layer width to interact with the univariate function at larger scales, which we leave to future work. The grid interval also makes QUARRY easier to inspect, since its behaviour can be read off per token. Combining QUARRY with methods that target the pruning threshold is a promising direction. A limitation of QUARRY is that the residual activation must be tuned separately for each model size.

6 Limitations

A limitation of QUARRY is that the layer width must be tuned separately for each model size. The layer width also makes QUARRY easier to inspect, since its behaviour can be read off per token. We expect the pruning threshold to interact with the grid interval at larger scales, which we leave to future work.

References

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